

Milestone 3: Full Dashboard Development – Visual Components Documentation

This documentation outlines the professional visualization suite developed for **Milestone 3** of the **Unified Military Analytics and Comparison Dashboard**. The objective of this milestone is to transform processed military and economic data into an interactive, four-module analytics ecosystem.

1. Module: Quick Stats (Global Overview)

The **Quick Stats** dashboard serves as the primary landing page, offering high-level insights into global military trends and rankings.

- **Global Power Rankings:** Visualizes the top 10 countries by **Power Index** to identify leading military forces.
- **Geospatial Analysis:** Includes a **Military Rank by Country** world map that allows users to visualize geographic concentrations of military power.
- **Economic & Asset Benchmarking:** Uses bar and bubble charts to compare top nations by **Purchasing Power Parity (PPP)**, **Total Military Assets**, and **Defense Budget (USD)**.
- **Dynamic KPI Cards:** Displays real-time values for critical metrics such as total global manpower or average defense spending.
- **Regional Filtering:** Provides navigation and filters for **Region, Continent, and Alliance** (e.g., NATO) to segment global data.

2. Module: Nation Overview (Individual Profile)

This module provides a deep-dive analysis of specific country capabilities, moving from global trends to granular data.

- **Comprehensive Country Profiles:** Users can select a specific country to display its full defense profile.
- **Multidimensional Metrics:** Utilizes **bar and radar charts** to visualize balanced capabilities across land, air, and sea power.
- **Contextual Tooltips:** Integrated tooltips provide immediate data on **Power Index Rank Gaps** and specific asset counts upon hovering.

3. Module: Compare Powers (Side-by-Side Analytics)

Designed for strategic benchmarking, this module enables direct head-to-head comparisons.

- **Dual-Country Selection:** Uses **Tableau parameters** to allow users to select any two nations for simultaneous comparison.

- **Comparative KPI Mapping:** Side-by-side visualization of **manpower, aircraft, navy, and defense budget**.
- **Economic Efficiency:** Compares derived KPIs such as the **Defense Budget-to-GDP Ratio** and **Assets per Capita** between the chosen nations.

4. Module: Coalition Builder (Alliance Simulation)

The **Coalition Builder** is an interactive simulation tool for evaluating the strength of military alliances.

- **Multi-Country Aggregation:** Allows users to select multiple countries via filters to view their **aggregated totals** for assets and budget.
- **Alliance Benchmarking:** Compares a custom-built coalition's total strength against a single reference country or an existing major alliance.
- **Strategic KPI Modeling:** Calculates the combined "Alliance Power Index" to simulate how joint forces impact global standings.

Technical Integration & Navigation

- **Seamless Navigation:** All four dashboards are interconnected via **navigation buttons** and **parameter-linked filters**.
- **Unified Ecosystem:** Filters applied in one module (e.g., Region) are linked across all tabs to ensure a consistent user experience.
- **Tool Stack:** Developed using **Tableau** and **Power BI**, with backend data derived from specialized Python KPI engineering.